But this does not happen. The experiments show that maximum kinetic energy of the photoelectrons emitted does not depend on the intensity of the incident light.

ii) According to wave theory, if a sufficiently intense beam of light is incident on the surface, electrons should be liberated from the surface of the target, however low the frequency of the radiation is.

From the experiments, it is found that photoelectric emission is not possible below a certain minimum frequency of incident radition. Therefore, the wave theory fails to explain the existence of threshold frequency.

iii) Since the energy of light is spread across the entire wavefront, the electrons which receive energy from it are large in number. Each electron needs considerable amount of time (a few hours) to get energy sufficient to overcome the work function and to get liberated from the surface.

But experiments show that photoelectric emission is almost instantaneous process (the time lag is less than 10^{-9} *s* after the surface is illuminated) which could not be explained by wave theory.

Thus, the experimental observations of photoelectric emission could not be explained on the basis of the wave theory of light.

EXAMPLE 8.1

For the photoelectric emission from cesium, show that wave theory predicts that

i) maximum kinetic energy of the photoelectrons (K_{max}) depends on the intensity I of the incident light

ii) K_{max} does not depend on the frequency of the incident light and

iii) the time interval between the incidence of light and the ejection of photoelectrons is very long.

For the sake of simplicity, the following standard assumptions can be made when light is incident on the given material.

a) Light is absorbed in the top atomic layer of the metal

b) For a given element, each atom absorbs an equal amount of energy and this energy is proportional to its cross-sectional area A

c) Each atom gives this energy to one of the electrons.

(Given : The work function for cesium is 2.14 eV and the power absorbed per unit area is $1.60\times10^{-6}\,\mathrm{Wm}^{-2}$ which produces a measurable photocurrent in cesium.)

Solution

i) According to wave theory, the energy in a light wave is spread out uniformly and continuously over the wavefront.

The energy absorbed by each electron in time t is given by

$$E = IAt$$

With this energy absorbed, the most energetic electron is released with K_{max} by overcoming the surface energy barrier or work function ϕ_0 and this is expressed as

$$K_{max} = IAt - \phi_0 \qquad (1)$$

Thus, wave theory predicts that for a unit time, at low light intensities when $IA < \phi_0$, no electrons are emitted. At higher intensities, when $IA \geq \phi_{0,}$ electrons are emitted. This implies that higher the light intensity, greater will be $K_{max.}$

K_{max} is dependent only on the intensity under given conditions – that is, by suitably increasing the intensity, one can produce

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